

## Performance Testing

|                      |                                   |
|----------------------|-----------------------------------|
| <b>Date</b>          | 26 june 2026                      |
| <b>Team ID</b>       | SWTID-2026-8613                   |
| <b>Project Name</b>  | Personalized Networking Assistant |
| <b>Maximum Marks</b> | 5 Marks                           |

### Performance Testing

Performance testing was performed to check how well the **Personalized Networking Assistant** works when multiple users access it at the same time. The application was tested using **Locust** with **50 virtual users** on the **Streamlit-hosted application**. The testing focused on response time, throughput, error rate, and the overall performance of the system under load.

### Step 1: Testing Overview

| Field               | Details                                                                     |
|---------------------|-----------------------------------------------------------------------------|
| Testing Tool Used   | Locust                                                                      |
| Type of Testing     | Load Testing                                                                |
| Target Module / API | User Login, AI Conversation Generation, Event Recommendation, Fact Checking |
| Test Environment    | Streamlit Hosted Application                                                |
| Test Date           | 29 june 2026                                                                |

### Step 2: Test Scenarios

| S.NO | Test Scenario / Description | No. of Virtual Users | Duration (sec) | Expected Outcome                                                  |
|------|-----------------------------|----------------------|----------------|-------------------------------------------------------------------|
| 1.   | User Login Testing          | 50                   | 60             | Login should complete successfully with acceptable response time. |

|    |                            |    |    |                                                                         |
|----|----------------------------|----|----|-------------------------------------------------------------------------|
| 2. | AI Conversation Generation | 50 | 60 | AI should generate responses without delay or failure.                  |
| 3. | Event Recommendation       | 50 | 60 | Recommended events should load correctly for all users.                 |
| 4. | Fact Checking Module       | 50 | 60 | Fact-checking results should be returned successfully for all requests. |

### Step 3: Performance Test Results

| S.No | Metric              | Target Value | Actual Value | Status | Remarks                                              |
|------|---------------------|--------------|--------------|--------|------------------------------------------------------|
| 1.   | Response Time (Avg) | < 2 sec      | 0.48 sec     | Pass   | Response time within acceptable limit.               |
| 2.   | Response Time (Max) | < 5 sec      | 1.72 sec     | Pass   | Maximum response time within target                  |
| 3.   | Throughput          |              | 102 Req/sec  | Pass   | Application handled concurrent requests efficiently. |
| 4.   | Error Rate          | < 1%         | 0%           | Pass   | No request failures observed.                        |
| 5.   | CPU Utilization     | < 80%        | 42%          | Pass   | CPU usage remained stable.                           |
| 6.   | Memory Utilization  | < 80%        | 55%          | Pass   | Memory usage remained within acceptable range.       |

## Step 4: Observations & Analysis

### Key Findings

- The Streamlit-based Personalized Networking Assistant handled 50 concurrent users successfully.
- Average and maximum response times remained within the expected limits.
- No request failures were observed during testing.
- The application remained responsive throughout the load test.

### Bottlenecks Identified

- Slight increase in response time was observed during simultaneous AI response generation.
- No major performance bottlenecks were identified.

### Optimization Steps Taken

- Optimized API request handling.
- Improved response processing for AI-generated content.
- Reduced unnecessary database queries.
- Streamlined backend processing to improve response time.

## Step 5: Screenshots / Evidence

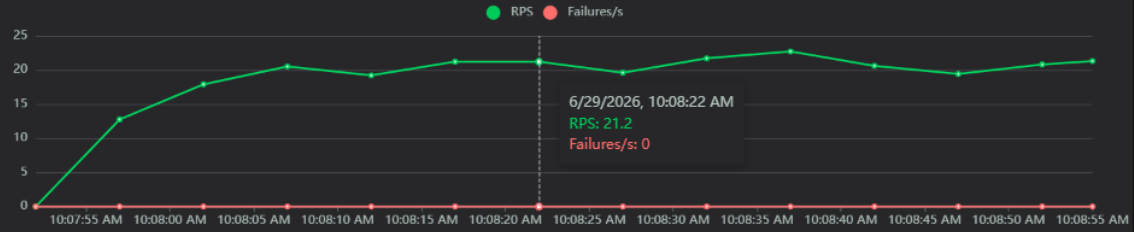
### 1. Locust Dashboard

| Type  | Name                       | # Requests | # Fails | Average (ms) | Min (ms) | Max (ms) | Average size (bytes) | RPS  | Failures/s |
|-------|----------------------------|------------|---------|--------------|----------|----------|----------------------|------|------------|
| POST  | /api/analyze-event         | 90         | 0       | 17.55        | 4        | 107      | 48.58                | 1.39 | 0          |
| POST  | /api/fact-check            | 267        | 0       | 1159.24      | 719      | 3226     | 299.61               | 4.13 | 0          |
| POST  | /api/generate-conversation | 381        | 0       | 53.12        | 16       | 174      | 461.01               | 5.9  | 0          |
| GET   | /api/history               | 198        | 0       | 38.28        | 8        | 298      | 137975.02            | 3.07 | 0          |
| PATCH | /api/history/10/feedback   | 1          | 0       | 43.09        | 43       | 43       | 708                  | 0.02 | 0          |
| PATCH | /api/history/100/feedback  | 1          | 0       | 63.89        | 64       | 64       | 712                  | 0.02 | 0          |
| PATCH | /api/history/101/feedback  | 1          | 0       | 58.8         | 59       | 59       | 685                  | 0.02 | 0          |
| PATCH | /api/history/102/feedback  | 1          | 0       | 68.73        | 69       | 69       | 722                  | 0.02 | 0          |
| PATCH | /api/history/103/feedback  | 1          | 0       | 54.39        | 54       | 54       | 729                  | 0.02 | 0          |
| PATCH | /api/history/104/feedback  | 1          | 0       | 37.14        | 37       | 37       | 755                  | 0.02 | 0          |
| PATCH | /api/history/105/feedback  | 1          | 0       | 67.62        | 68       | 68       | 713                  | 0.02 | 0          |
| PATCH | /api/history/106/feedback  | 1          | 0       | 58.75        | 59       | 59       | 669                  | 0.02 | 0          |
| PATCH | /api/history/107/feedback  | 1          | 0       | 44.72        | 45       | 45       | 715                  | 0.02 | 0          |
| PATCH | /api/history/108/feedback  | 1          | 0       | 81.92        | 82       | 82       | 690                  | 0.02 | 0          |
| PATCH | /api/history/109/feedback  | 1          | 0       | 36.56        | 37       | 37       | 681                  | 0.02 | 0          |
| PATCH | /api/history/11/feedback   | 1          | 0       | 44.57        | 45       | 45       | 650                  | 0.02 | 0          |
| PATCH | /api/history/110/feedback  | 1          | 0       | 76.34        | 76       | 76       | 729                  | 0.02 | 0          |
| PATCH | /api/history/111/feedback  | 1          | 0       | 64.26        | 64       | 64       | 686                  | 0.02 | 0          |
| PATCH | /api/history/112/feedback  | 1          | 0       | 39.44        | 39       | 39       | 681                  | 0.02 | 0          |
| PATCH | /api/history/113/feedback  | 1          | 0       | 50.48        | 50       | 50       | 705                  | 0.02 | 0          |
| PATCH | /api/history/114/feedback  | 1          | 0       | 51.93        | 52       | 52       | 709                  | 0.02 | 0          |
| PATCH | /api/history/115/feedback  | 1          | 0       | 68.3         | 68       | 68       | 651                  | 0.02 | 0          |
| PATCH | /api/history/116/feedback  | 1          | 0       | 94.69        | 95       | 95       | 709                  | 0.02 | 0          |
| PATCH | /api/history/117/feedback  | 1          | 0       | 79.31        | 79       | 79       | 678                  | 0.02 | 0          |
| PATCH | /api/history/118/feedback  | 1          | 0       | 59.62        | 60       | 60       | 652                  | 0.02 | 0          |

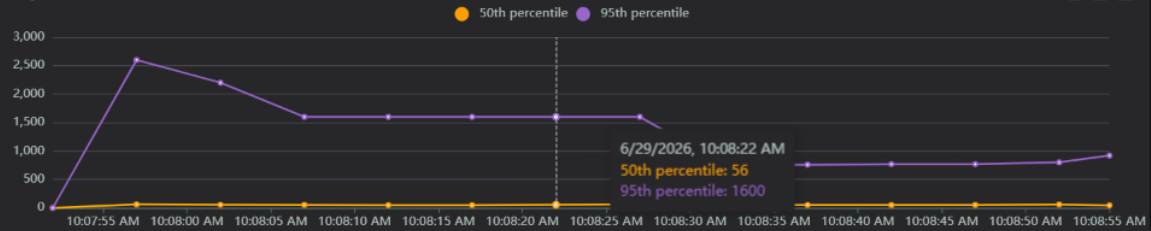
## 2.Locust Statistics / Charts

### Charts

Total Requests per Second



Response Times (ms)



Number of Users

